

ProblemsWeek2

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1 Geophysical Inversion

1.1 Homework 2

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1.2.1 Problem 1

What is the mean and variance of the uniform distribution $p(d) = 1$ on the interval $[0, 1]$

Solution The mean is $\langle x \rangle = E(x) = \int_{-\infty}^{\infty} xp(x)dx = \int_0^1 xdx = 0.5$ (It's a triangle that is half a unit square, no need to do calculus)

The variance is $\sigma^2 = \int_{-\infty}^{\infty} (x - \langle x \rangle)^2 p(x)dx = \int_0^1 (x - 0.5)^2 dx = \int_{-0.5}^{0.5} u^2 du = \frac{0.5^3 - (-0.5)^3}{3} = \frac{1}{12}$

with $u = x - 0.5$

1.2.2 Problem 3

Write a program to create a vector $\mathbf{d}$ of $N = 1000$ realisations of a Gaussian-distributed random variable with mean $\langle x \rangle = 4$ and variance $\sigma_x^2 = 2^2$

Count up the number of realisations of the variable where $x_i > (\langle x \rangle + 2\sigma_x)$ Is it about what you expected?

Solution

```
[2]: import numpy as np

def calculate_value(mean, stdev, size):
    d = np.random.normal(loc = mean, scale = stdev, size = size)

    ref = mean + 2*stdev
    mask = d > ref #Array of True/False for the boolean check
    return np.sum(mask) #True/False are just 1/0 under the hood so we can sum
    ↪ up the array to get a count

mean = 4
stdev = 2
size = 1000
```

```
calculate_value(mean, stdev, size)
```

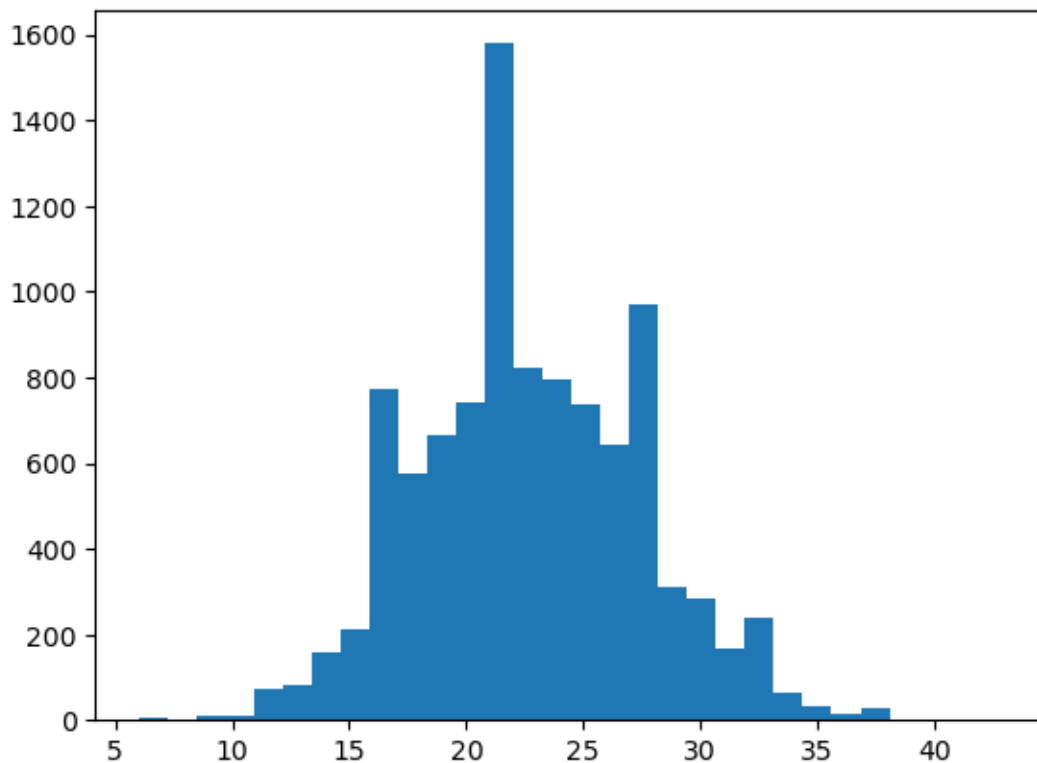
```
[2]: np.int64(18)
```

This is about what we expect. For a Gaussian-distributed variable about 95% of realisations should land within 2 standard deviations of the mean. Since we are only looking at realisations above the mean we expect around 2.5% of the 1000 realisations to be above the threshold. We can check this better by repeating the experiment

```
[3]: from matplotlib.pyplot import hist
    %matplotlib inline

    out = [calculate_value(4,2,1000) for _ in range(10000)]
    outvec = np.array(out)

    plot = hist(outvec, bins=30)
```



Indeed this is roughly what we should expect.

1.2.3 Problem 5

Use the transformation method to compute realizations of the probability density function $q(m) = 3m^2$ on the interval $(0,1)$, starting from realizations of the uniform distribution $p(d) = 1$. Check

your results by plotting a histogram.

Solution We calculate $d = \int q(m)dm = \int 3m^2 dm = m^3 = Q(m)$ We then invert to get $m = \sqrt[3]{d}$

```
[4]: def transform(N):  
      d = np.random.uniform(size = N)  
      m = np.cbrt(d)  
      return m  
  
plot = hist(transform(10000), bins = 30)
```

